



An Approach Combining SysML and Modelica for Modelling and Validate Wireless Sensor Networks

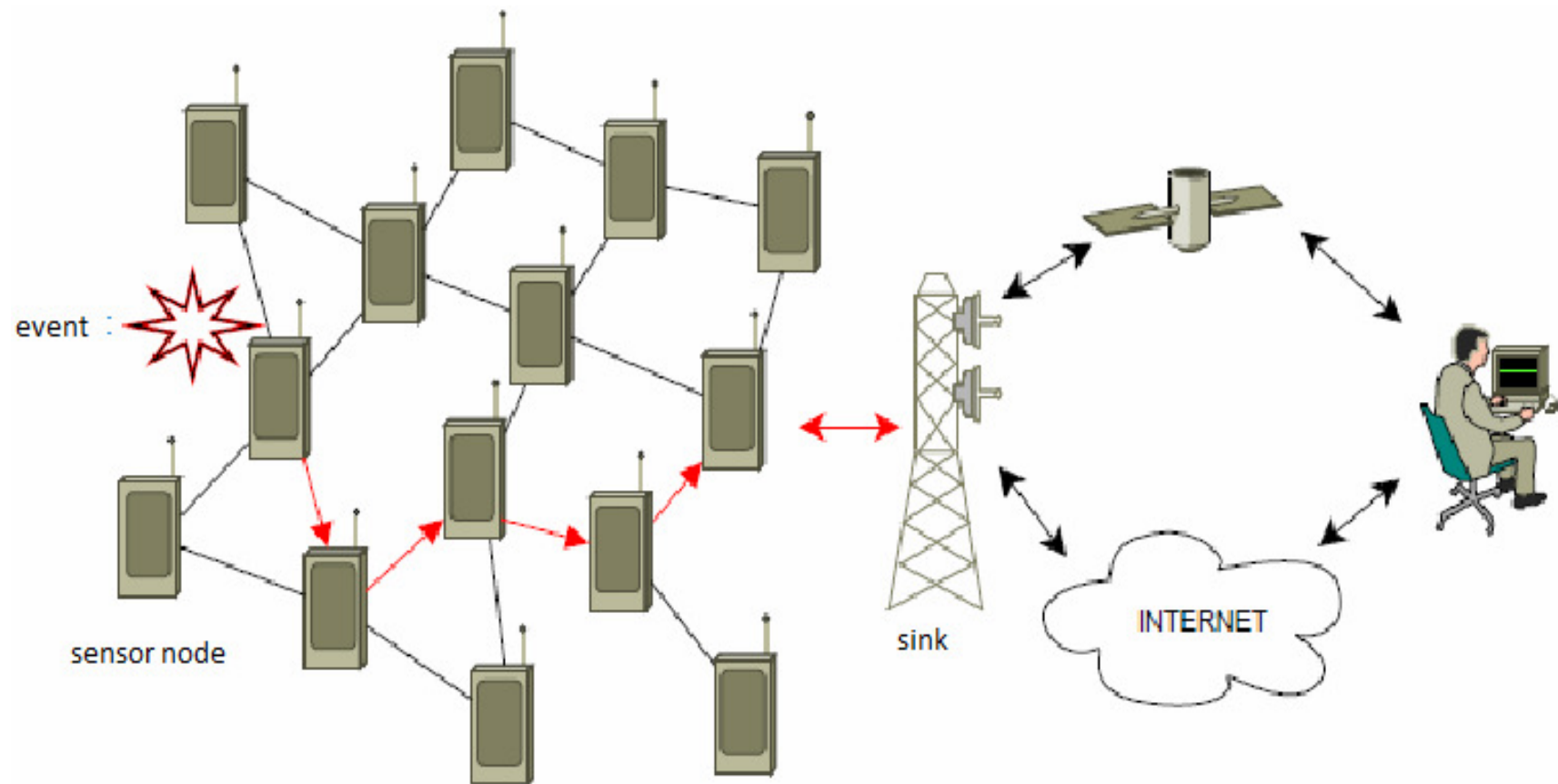
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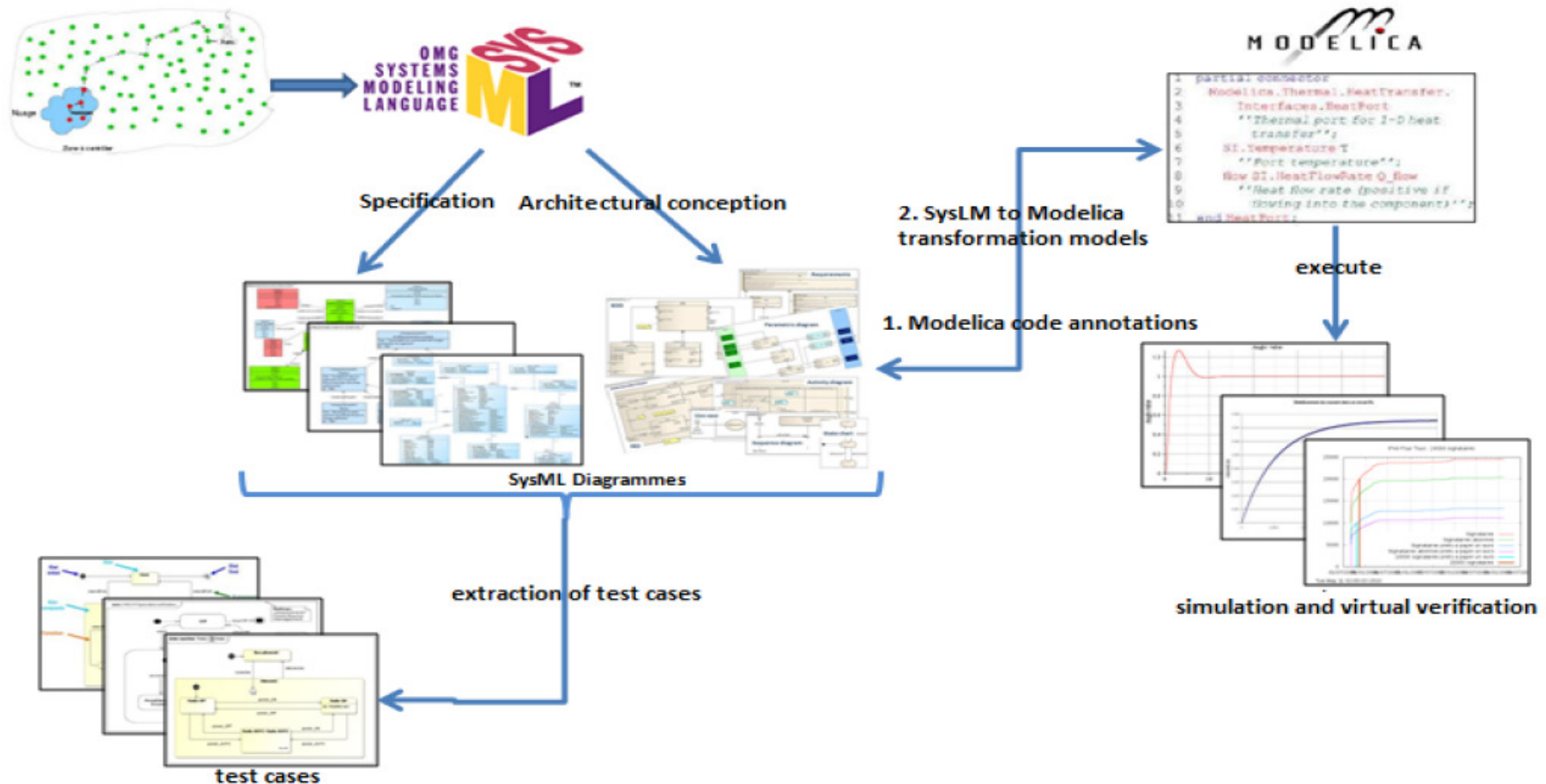
Outline

- Introduction
- Proposed Approach
- From SysML to Modelica
- Illustration
- Conclusion and Future work

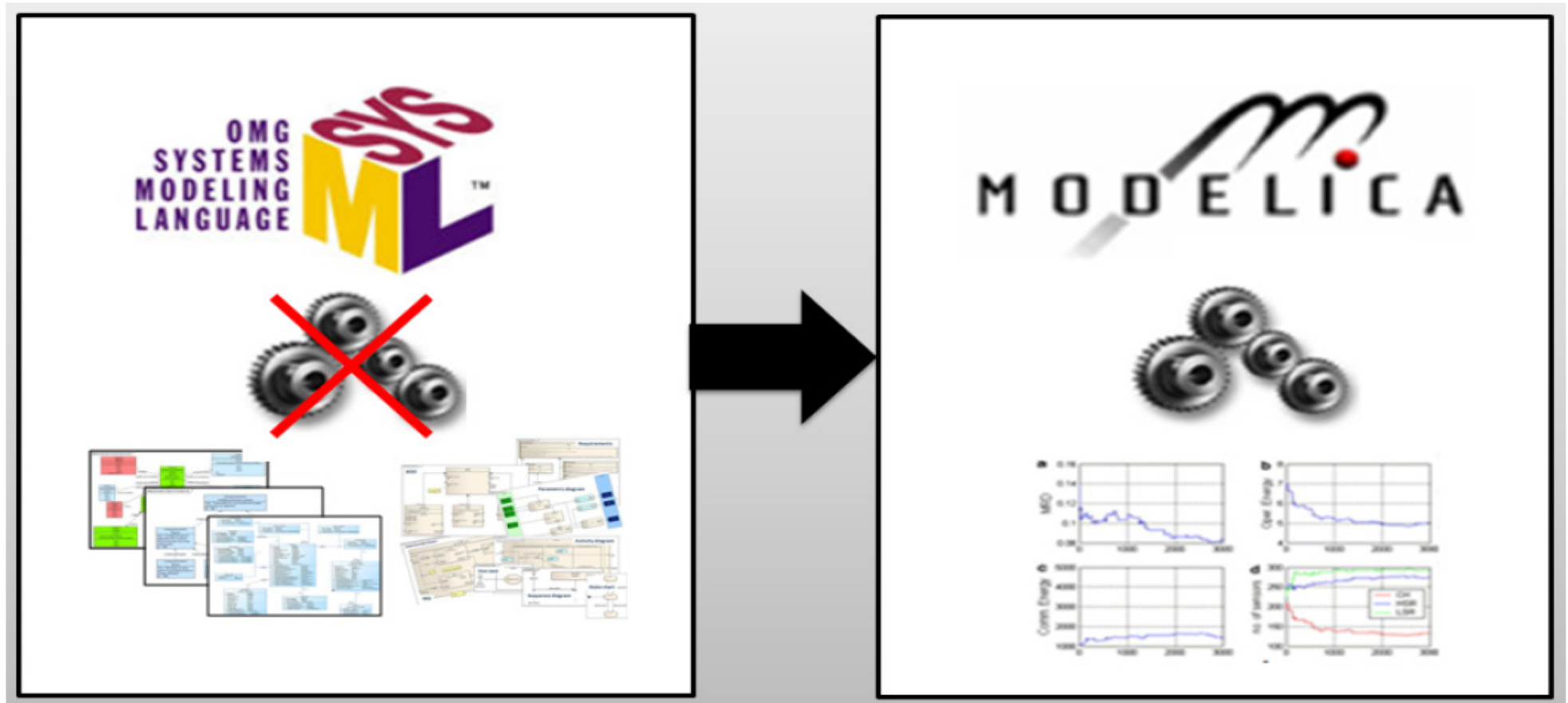
WSN Introduction



Proposed Approach



SysML/Modelica

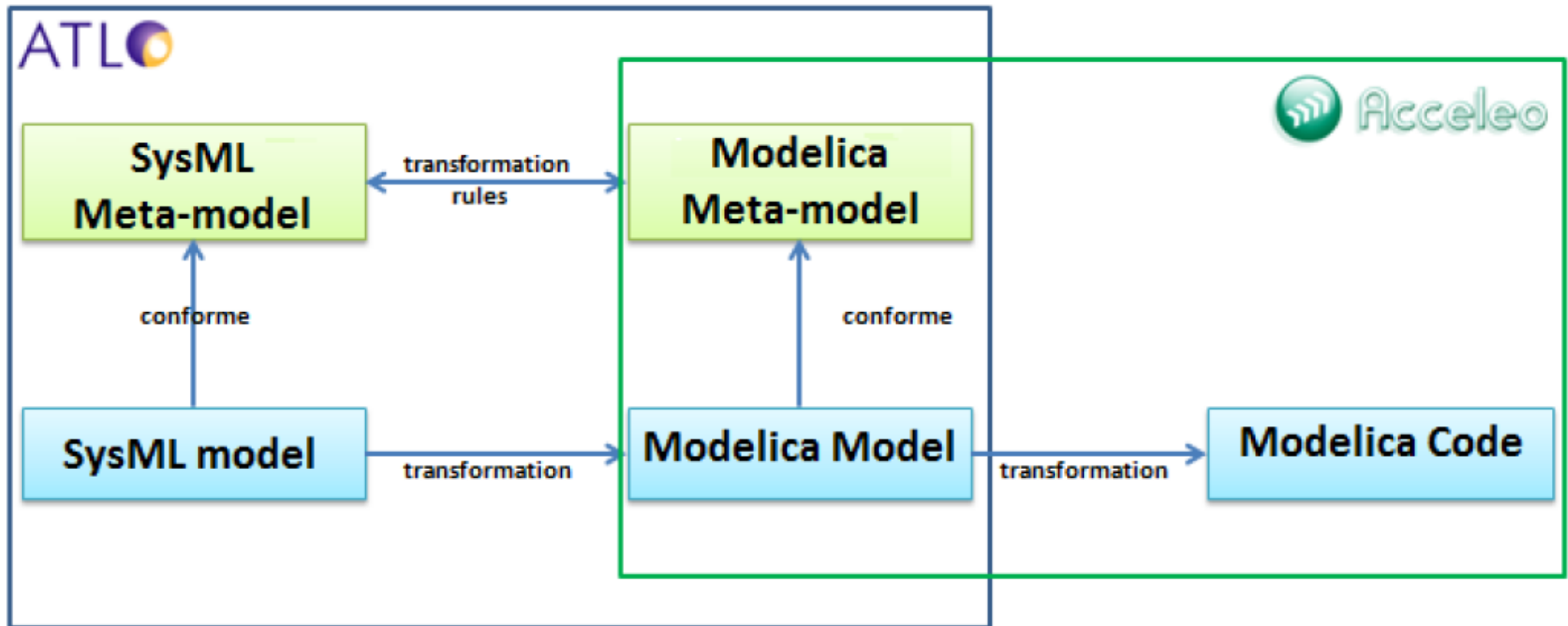


Modelica

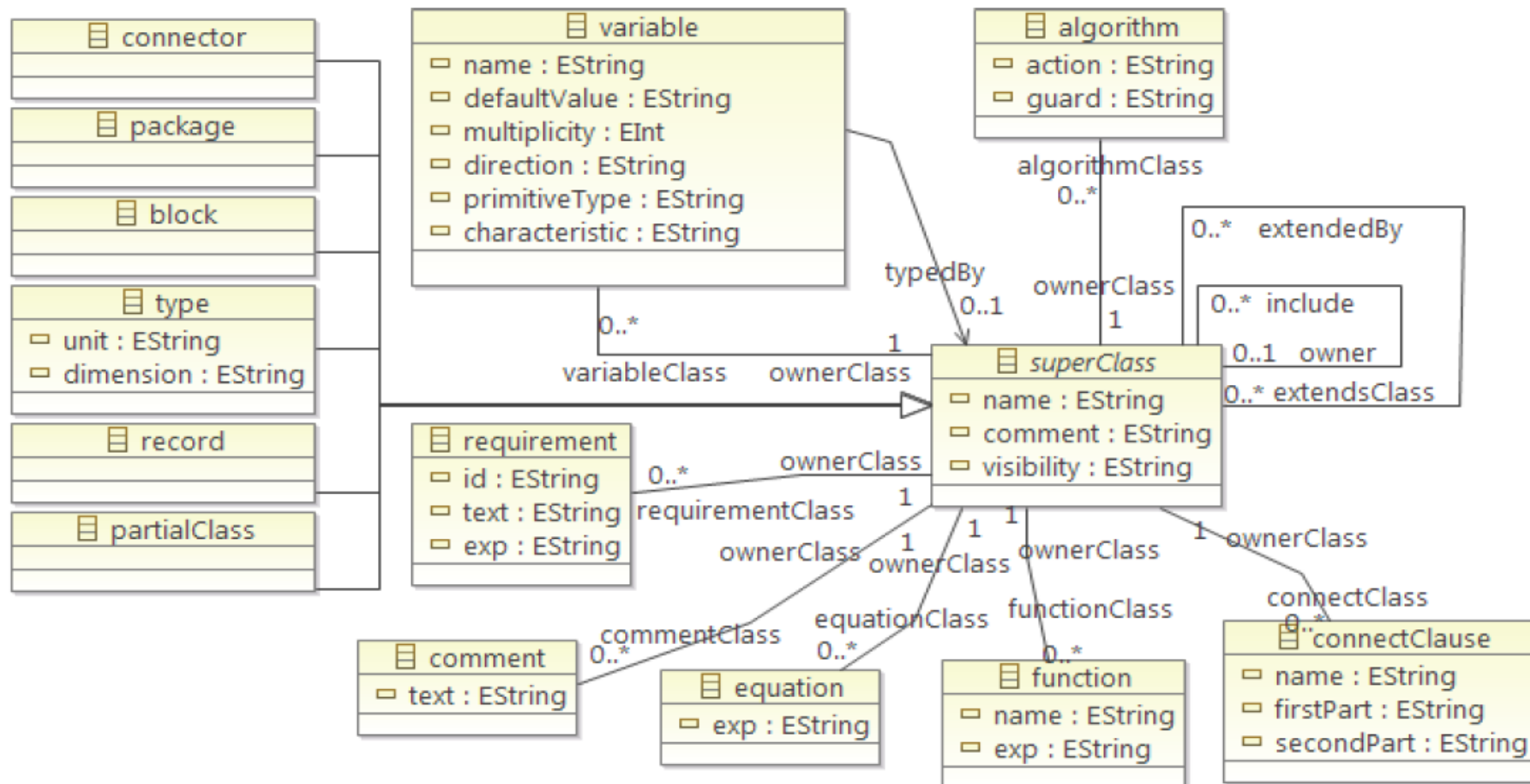


- Open source language
- Allows an object-oriented modeling approach, a graphical representation and simulation;
- Considers at any time the equation set which describe the model as single system equations;
- Modelica compiler has single simulator that allows automatic synchronization between discrete and continuous parts.

Model Transformation Tools



Modelica Meta-model

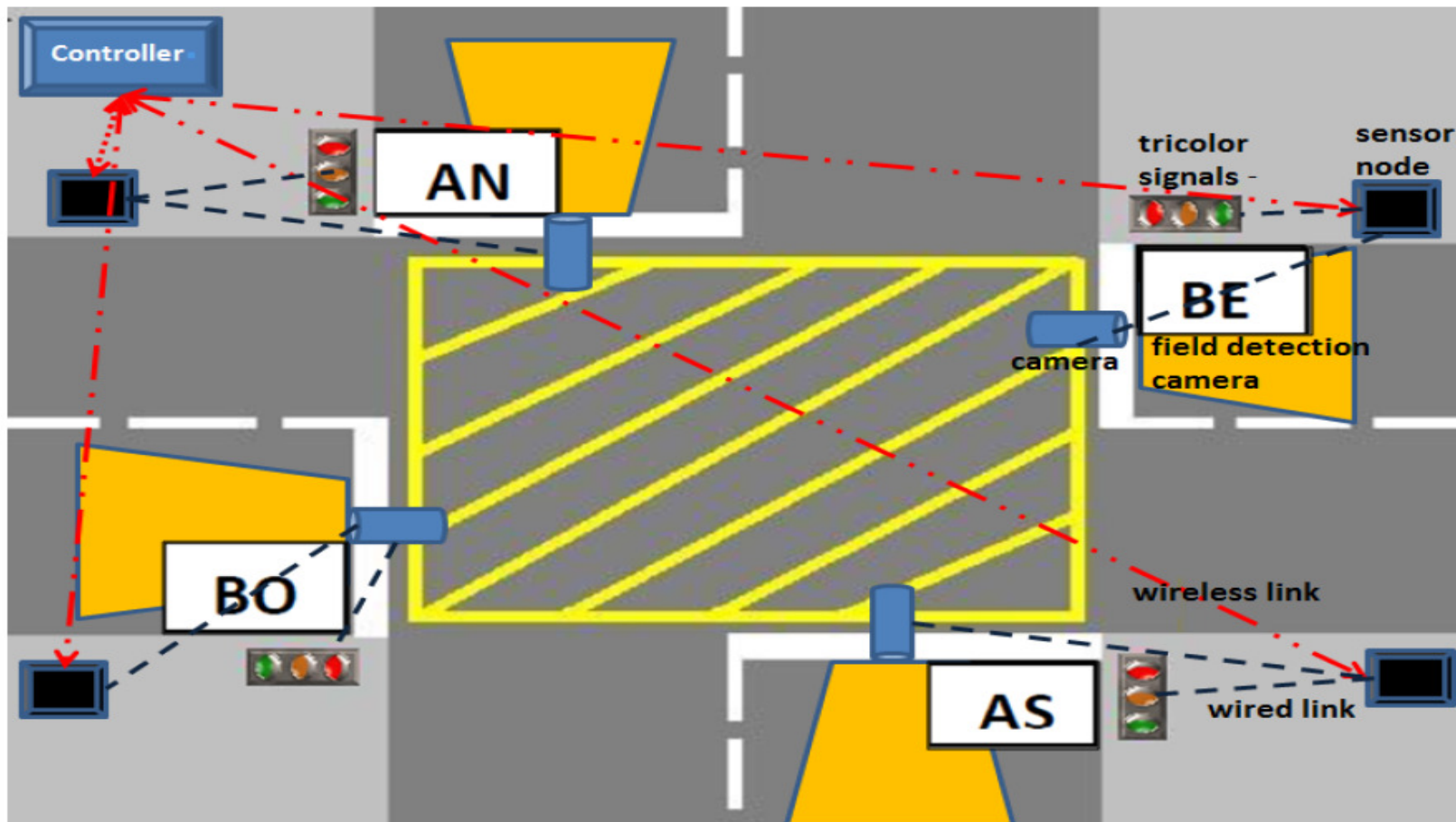


SysML to Modelica

Transformation rules

SysML Model	Modelica Model
package, block, abstract-block	package, block, partial-class
flow-specification, value-type	connector, type
flow-property, flow-port	property, connector
connector flux(x,y)	equation connect(x,y)
constraint property	equation
state machine guard	'when' statement guard
operation without a value to return	instruction block of 'when' statement
operation with a value to return	function
requirement	boolean expression (invariant or safety constraint)

Case study: Crossroads system



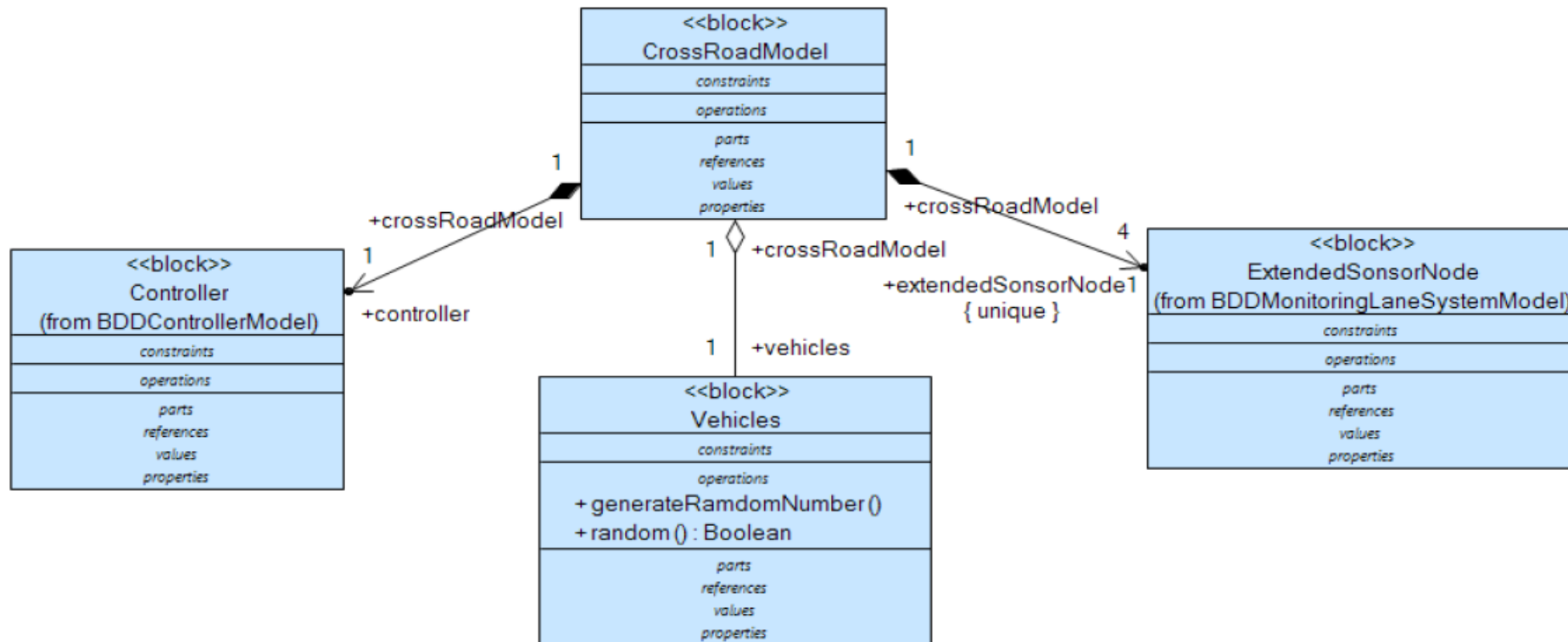
Adopted Energy Model

$$E_{SensorNode} = (E_{Sensing} + E_{Processing} + E_{Transmitting} + E_{Receiving} + E_{Transient} + E_{Logging})^1$$

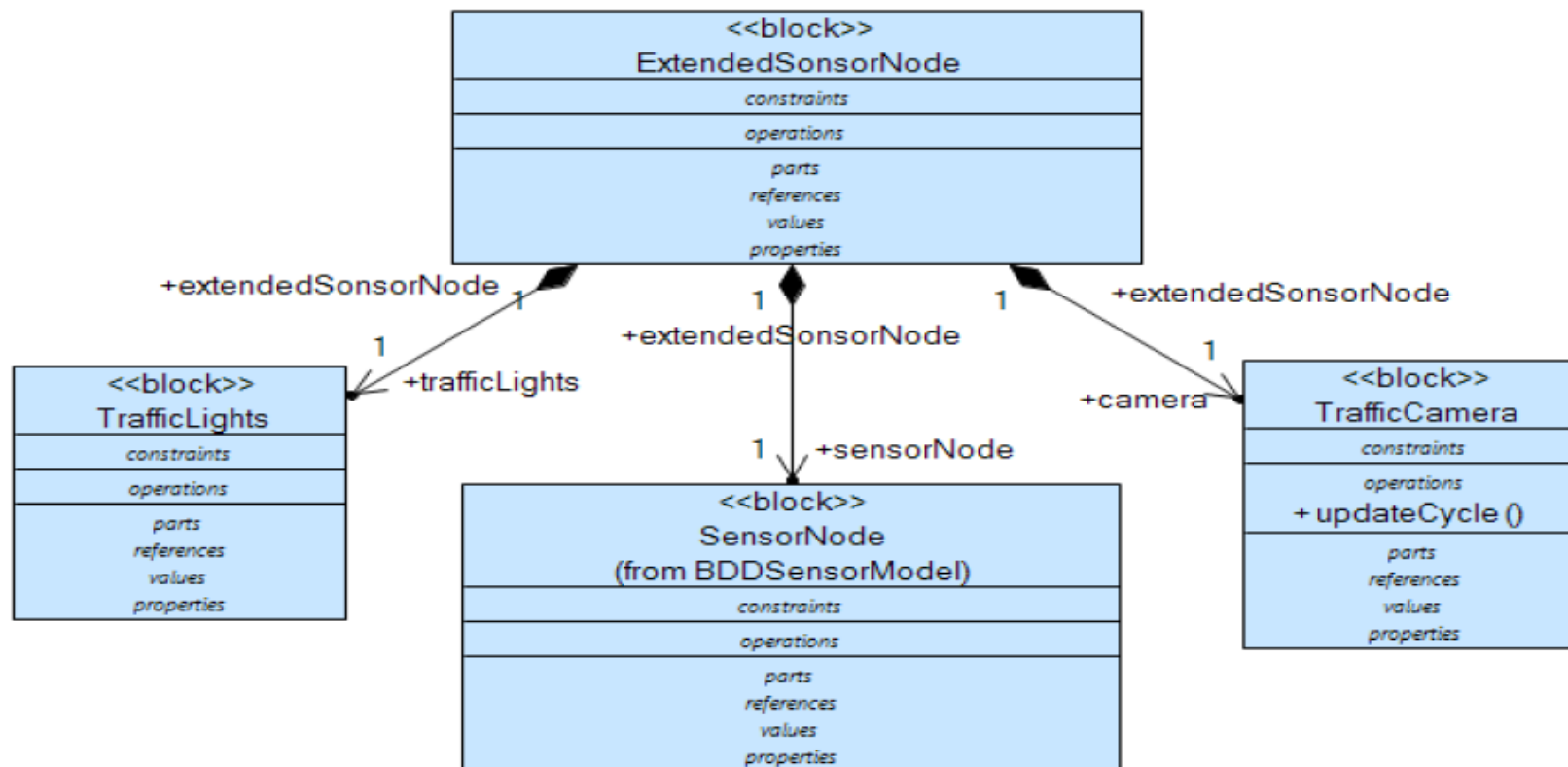
- ▶ $E_{Sensing} = bV_{sup}I_{sens}T_{sens}$
- ▶ $E_{Processing} = E_{switch} + E_{leak}$;
- ▶ $E_{Transmitting} = bE_{elec} + bd_{ij}^n E_{amp}$
- ▶ $E_{Receiving} = bE_{elec}$
- ▶ $E_{Transient} = T_A V_{sup} [c_N I_A + (1 - c_N) I_S]$
- ▶ $E_{Logging} = E_{write} + E_{read}$

¹ : M. N. Halgamuge, M. Zukerman, K. Ramamohanarao, H. L. Vu. An estimation of sensor energy consumption.

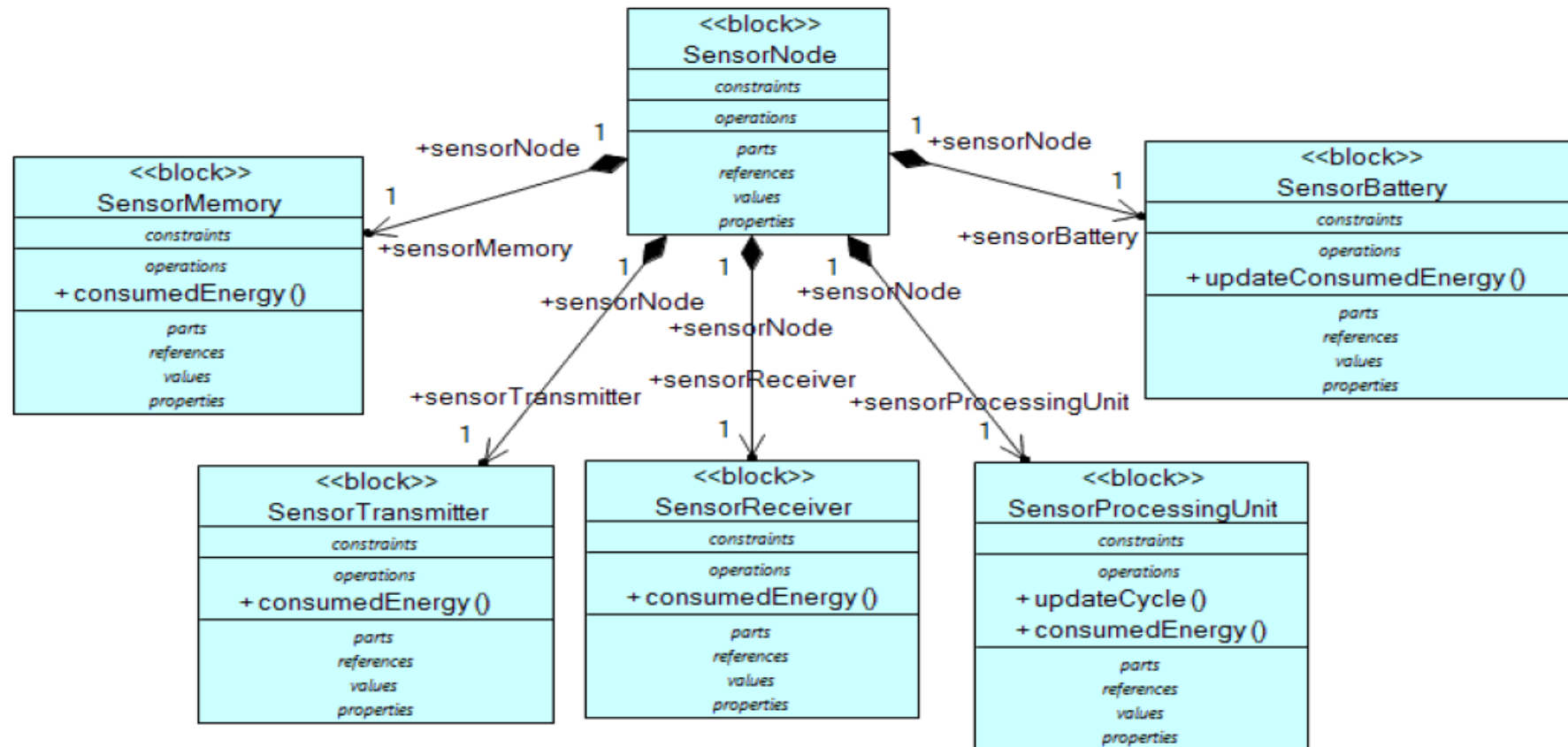
BDD of the Global System



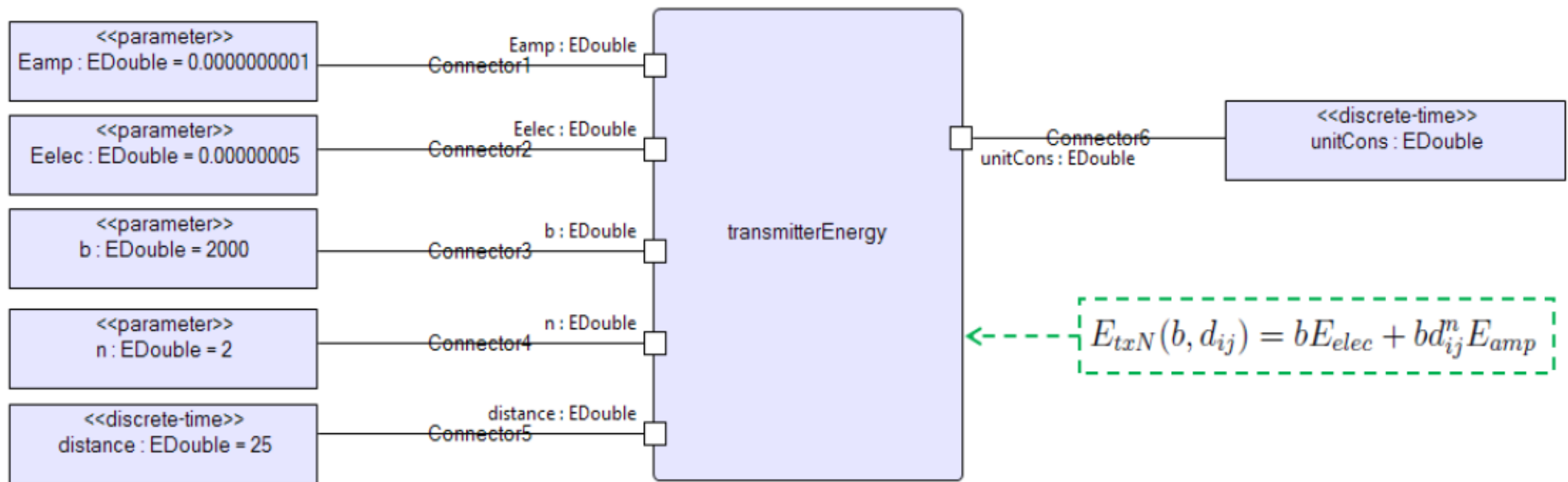
BDD of the extended Sensor



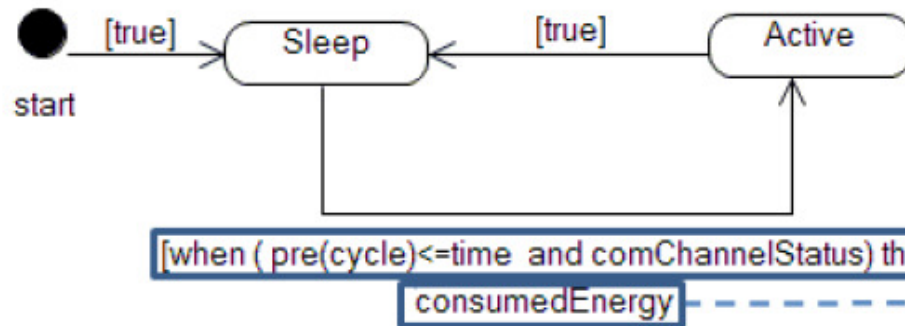
BDD of the simple Sensor



Transmitter Parametric Diagram

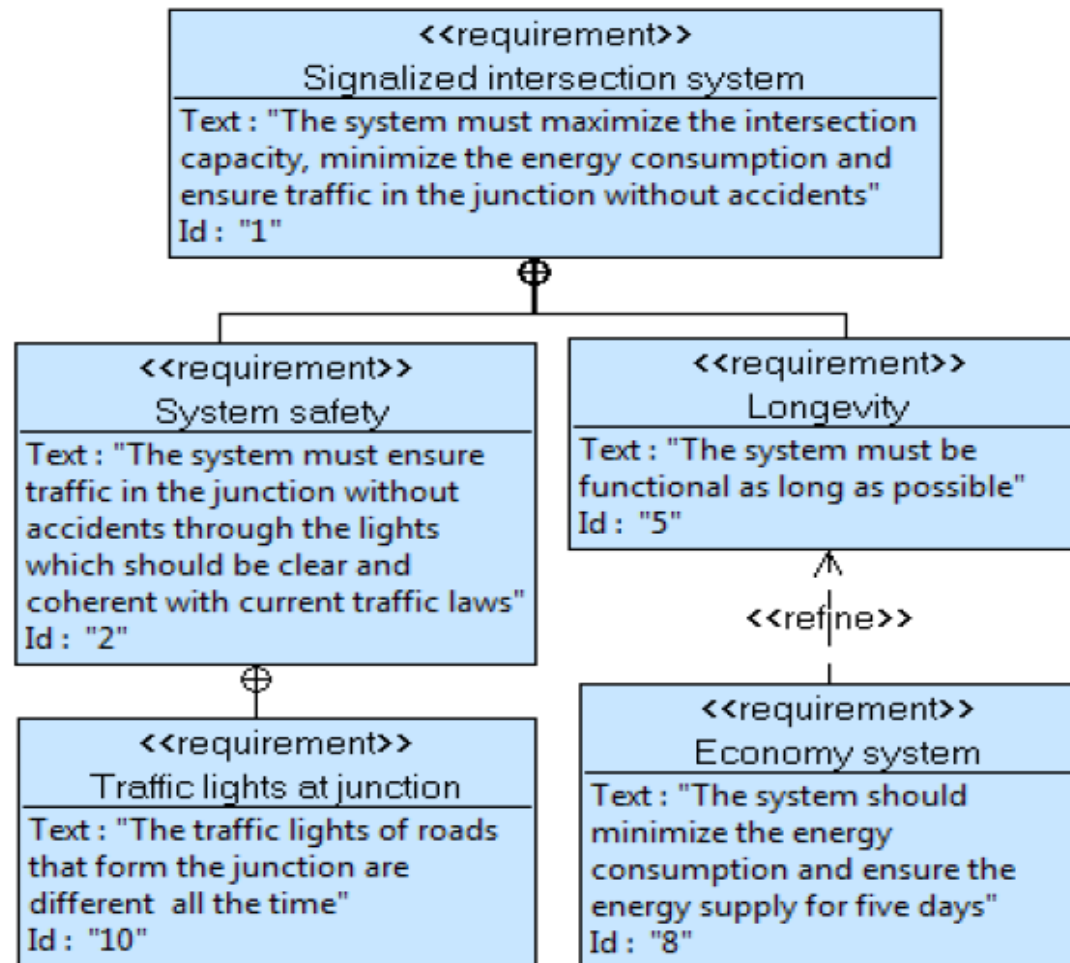


Transmitter State Machine Diagram

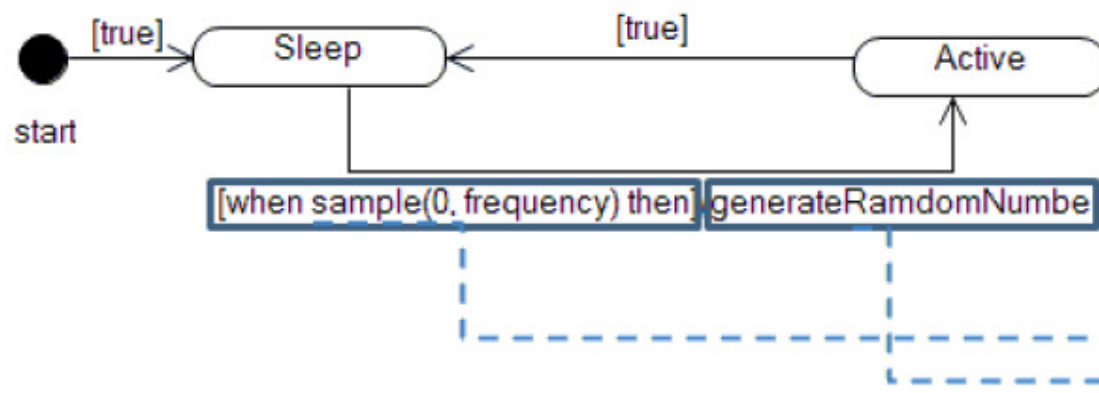


```
1 block SensorTransmitter
2   ...
3   Real distance(start = 25);
4   Real unitCons(start = 0);
5   Real cycle(start = 0.01);
6   ...
7   equation
8     ...
9   algorithm
10    when pre(cycle) <= time and comChannelStatus then
11      consumedEnergy:=pre(consumedEnergy) + unitCons;
12      ...
13    end when;
14 end SensorTransmitter;
```

Requirement Diagrams

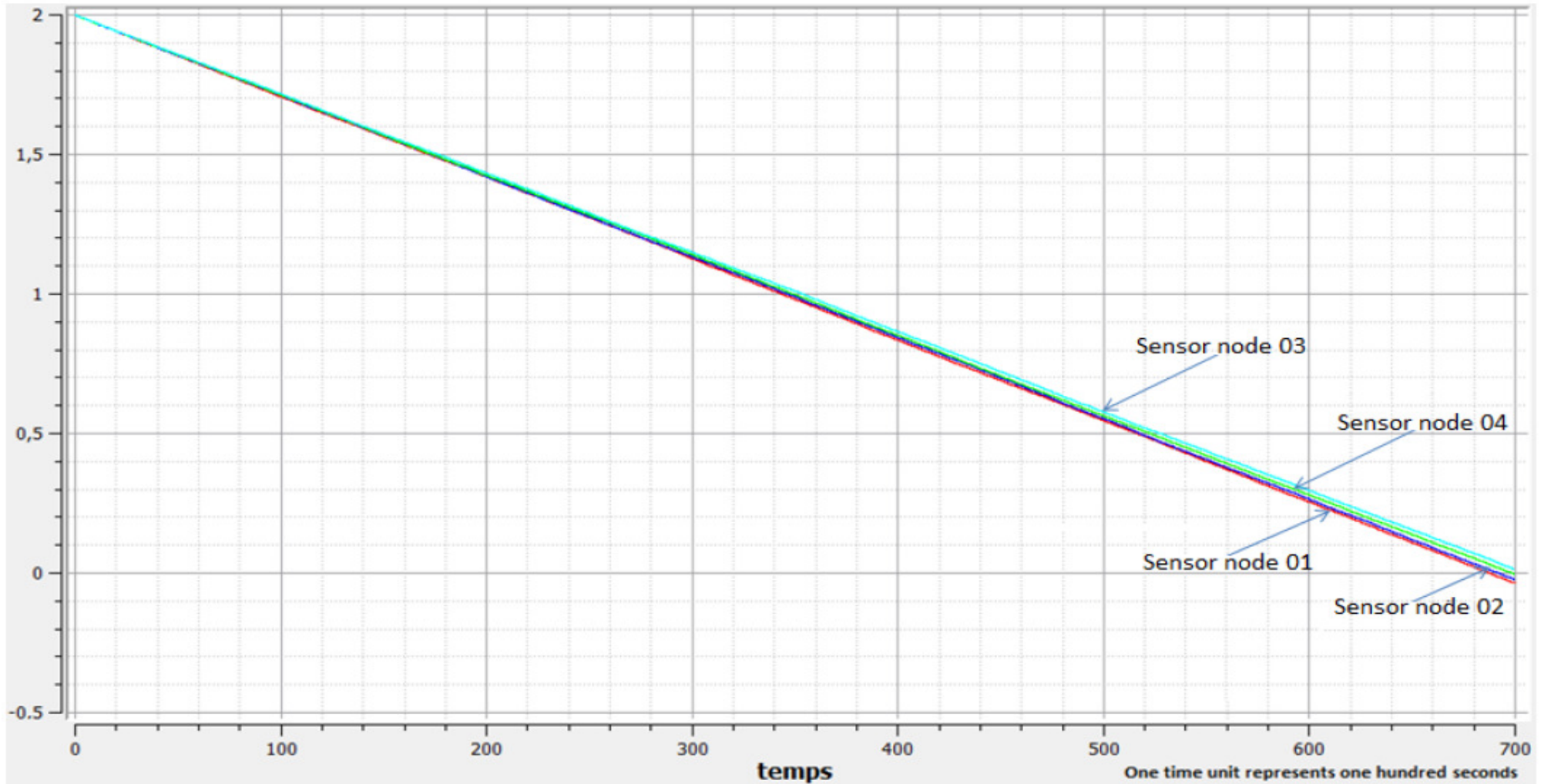


Random Scenario

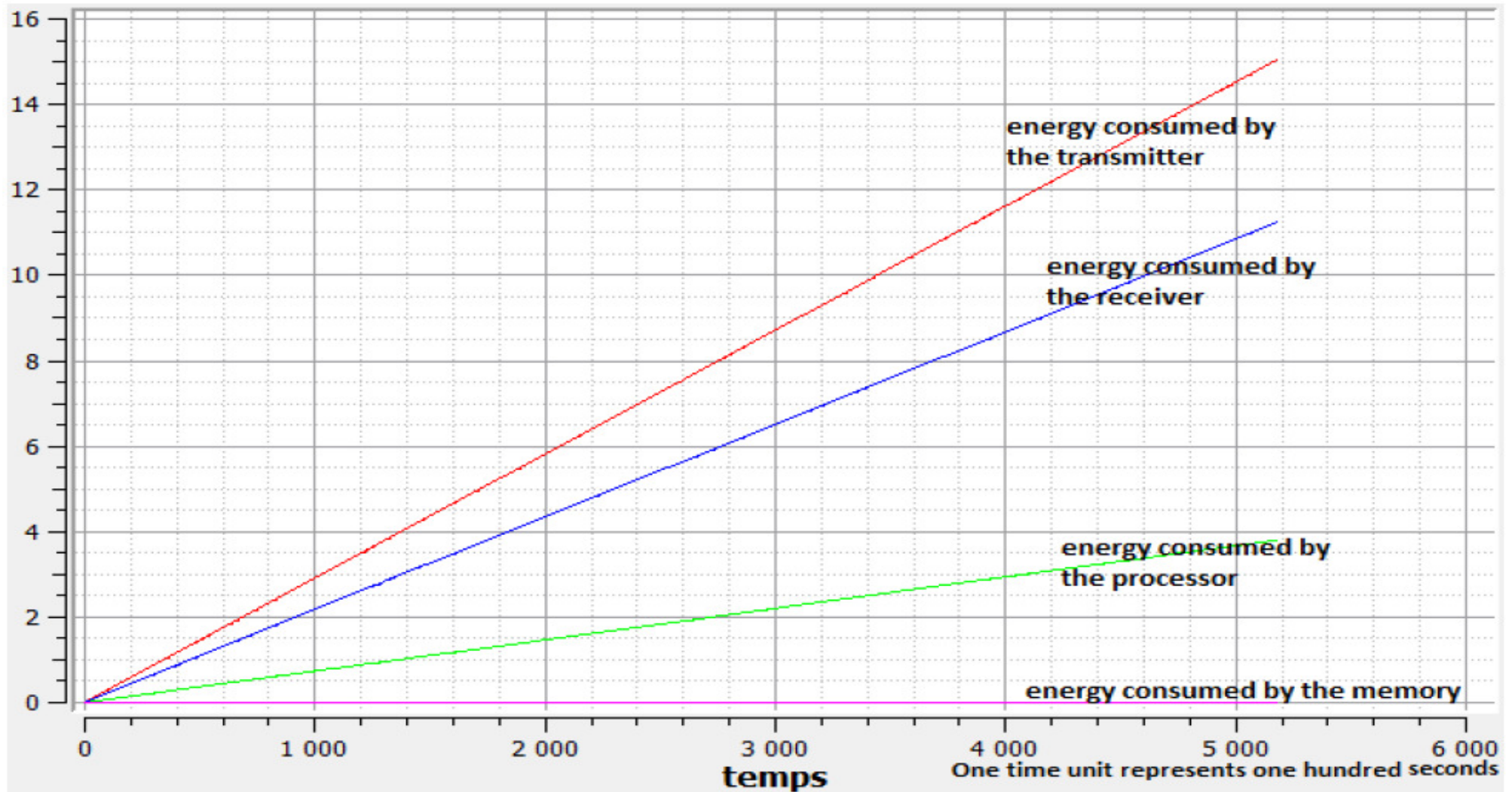


```
1 block Vehicles
2   ...
3   parameter Real frequency = 0.05;
4   output Data nbrLane12;
5   output Data nbrLane21;
6   output Data nbrLane22;
7   output Data nbrLane11;
8   algorithm
9     when sample(0, frequency) then
10      (nbrLane11.value, seed1) := random(pre(seed1));
11      (nbrLane12.value, seed2) := random(pre(seed2));
12      (nbrLane21.value, seed3) := random(pre(seed3));
13      (nbrLane22.value, seed4) := random(pre(seed4));
14    end when;
15 end Vehicles;
```

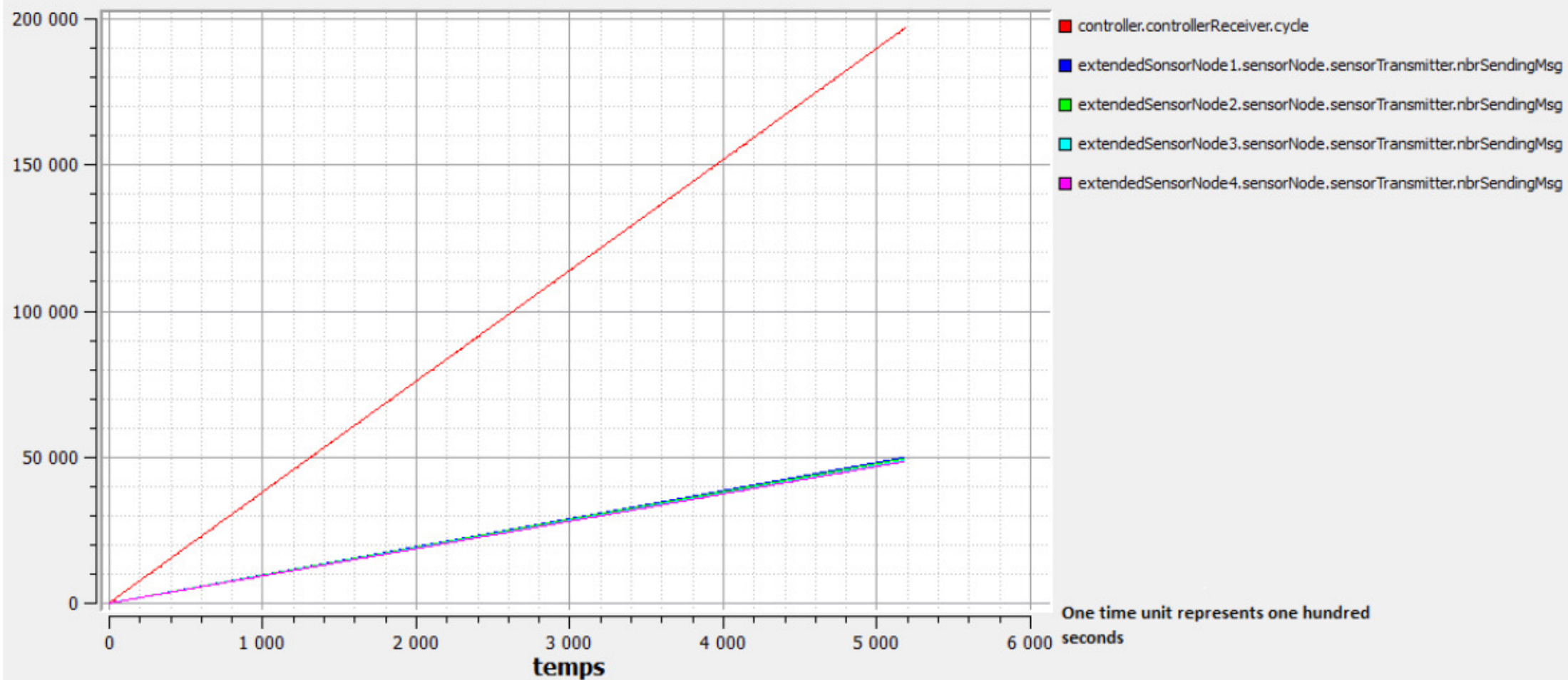
System Energy Consumption



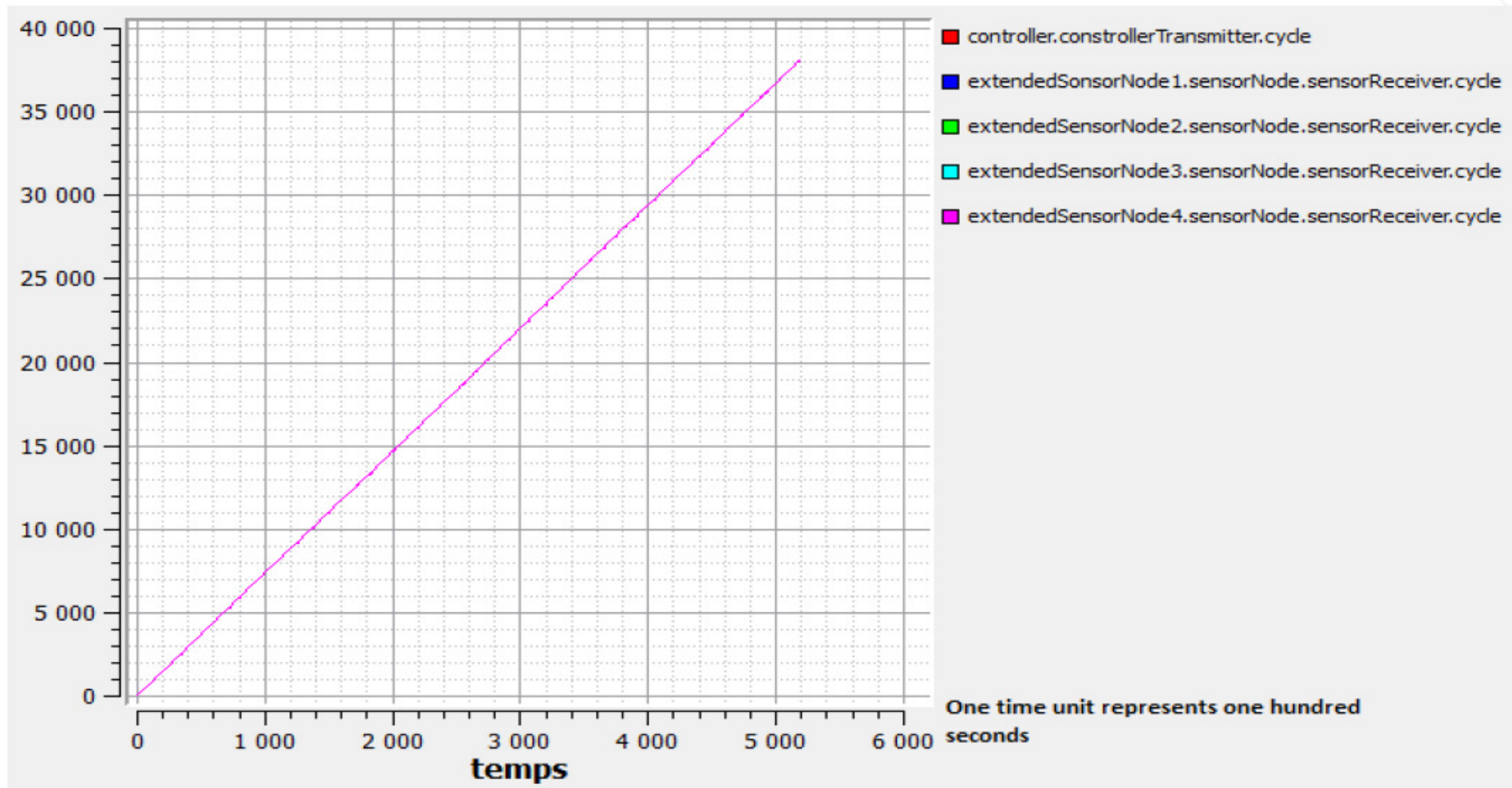
Sensor node Energy Consumption



Messages received by the Controller



Messages sent by the Controller



Conclusion

- Approach to design WSN using the MDA standard;
- Set of rules to transform the SysML model to the Modelica model taking into account static, dynamic and requirement diagrams of SysML;
- A virtual verification/simulation and a requirement tracing of WSN.

Future work



- Use Sequence and Activity Diagrams to describe the actions of State Machine Diagrams;
- Validate the correctness of transformation rules;
- Verification of functional properties;
- Investigate the automatic reconfiguration property of WSN.

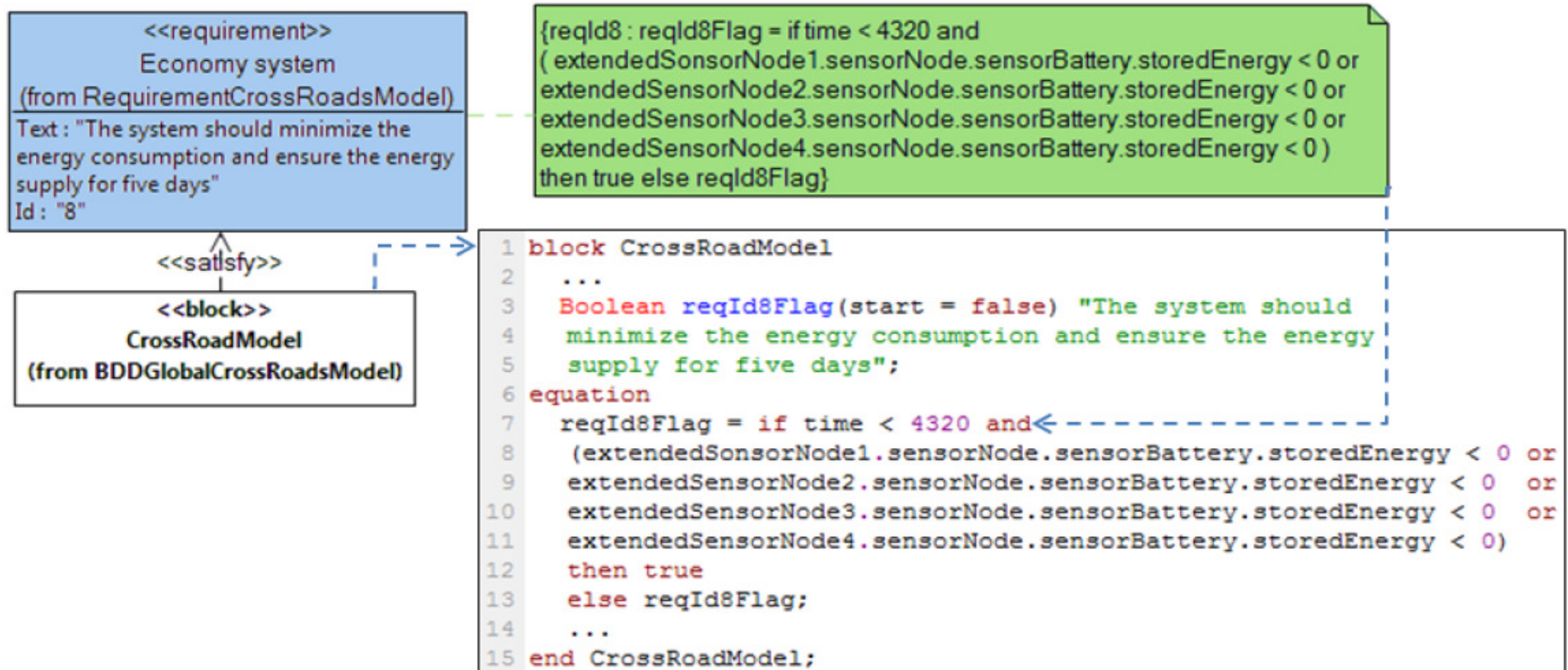


Tank you for your Attention

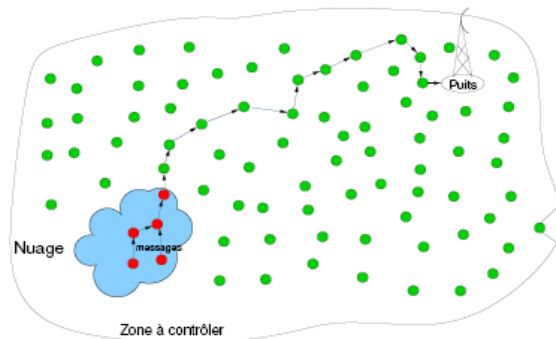
Context/problematic



Non-functional property

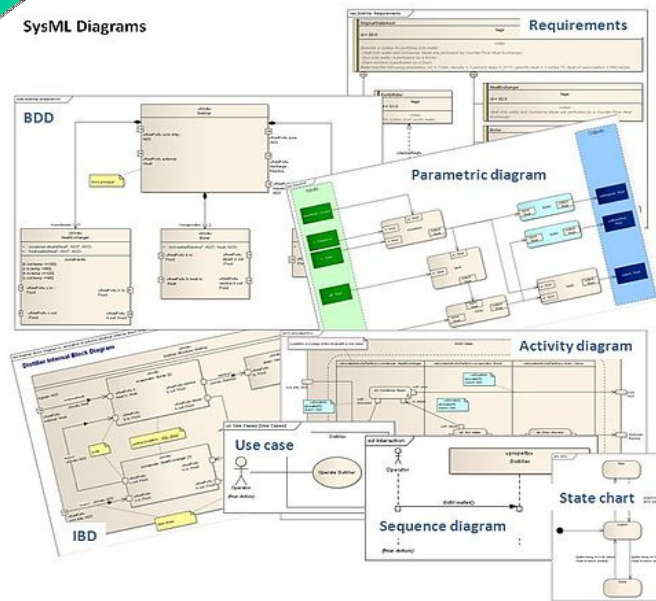


Approach



Specification

SysML Diagrams



Requirements
(Propertiess)

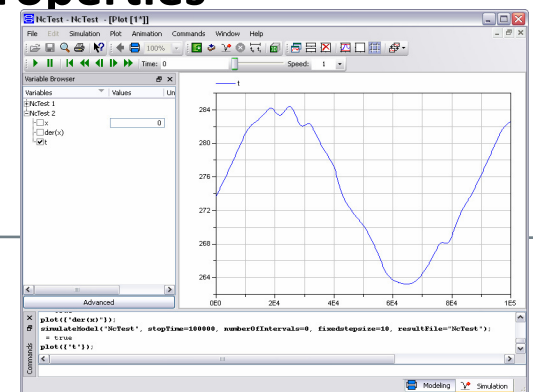
Vérification of functional
properties



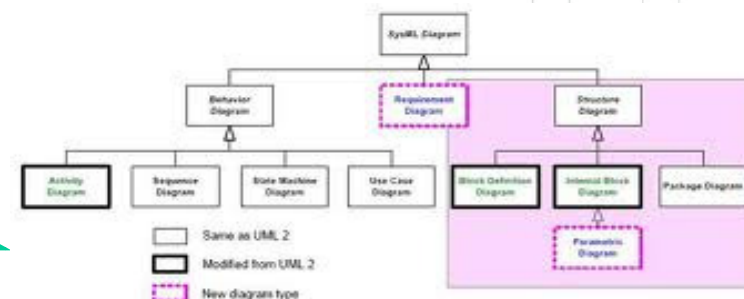
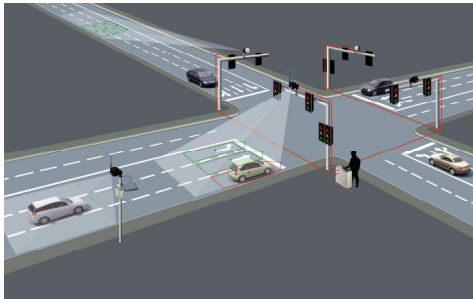
```
1 partial connector
2   Modelica.Thermal.HeatTransfer.
3     Interfaces.HeatPort
4     "Thermal port for 1-D heat
5     transfer";
6   SI.Temperature T
7     "Port temperature";
8   flow SI.HeatFlowRate Q_flow
9     "Heat flow rate (positive if
10    flowing into the component)";
11 end HeatPort;
```

Transformation

Simulation of non-functional
properties



Case study

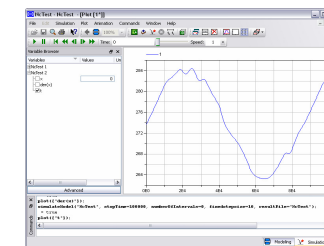
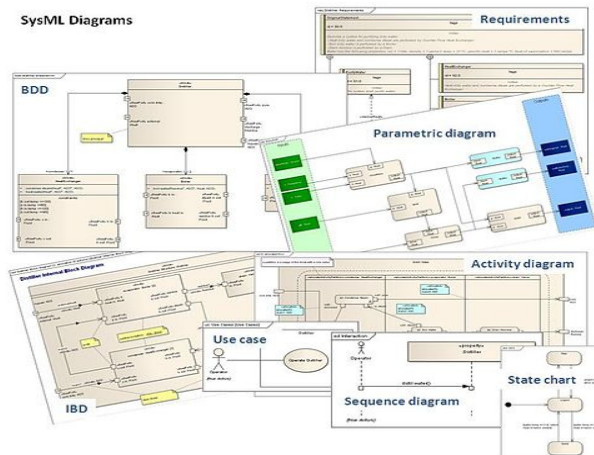


MODELICA

```

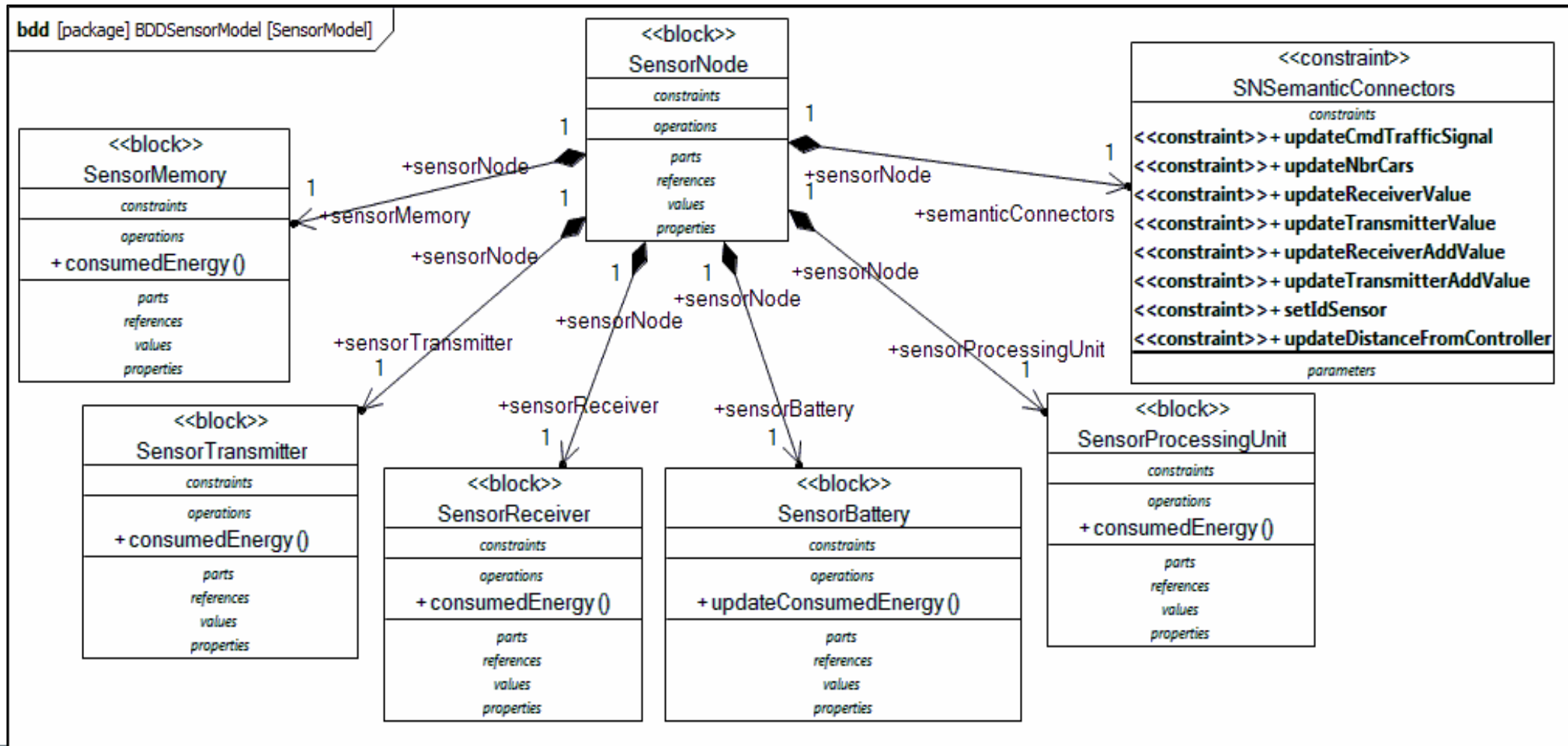
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11 end HeatPort;
    
```

Assert(□ Energy>0)

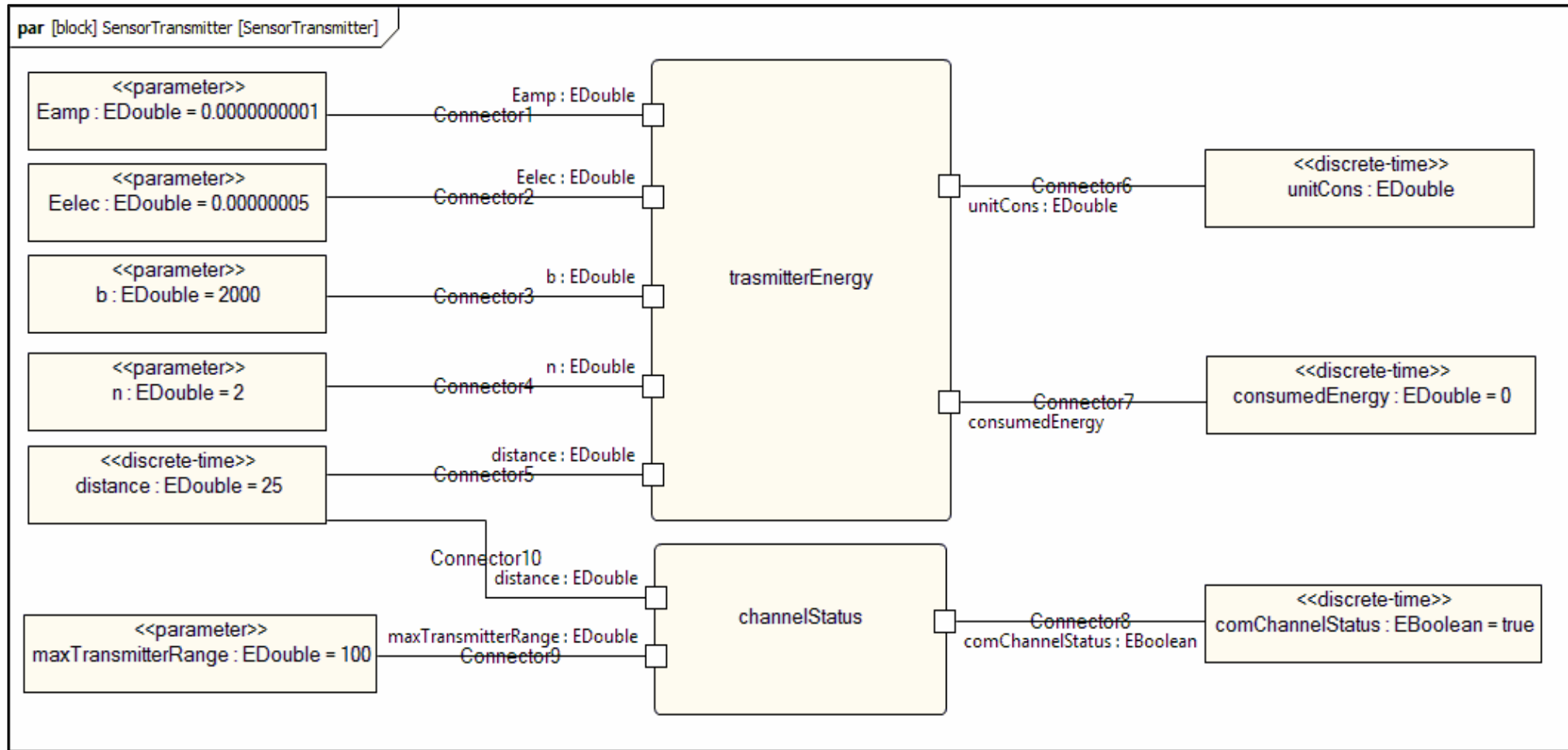


Verification

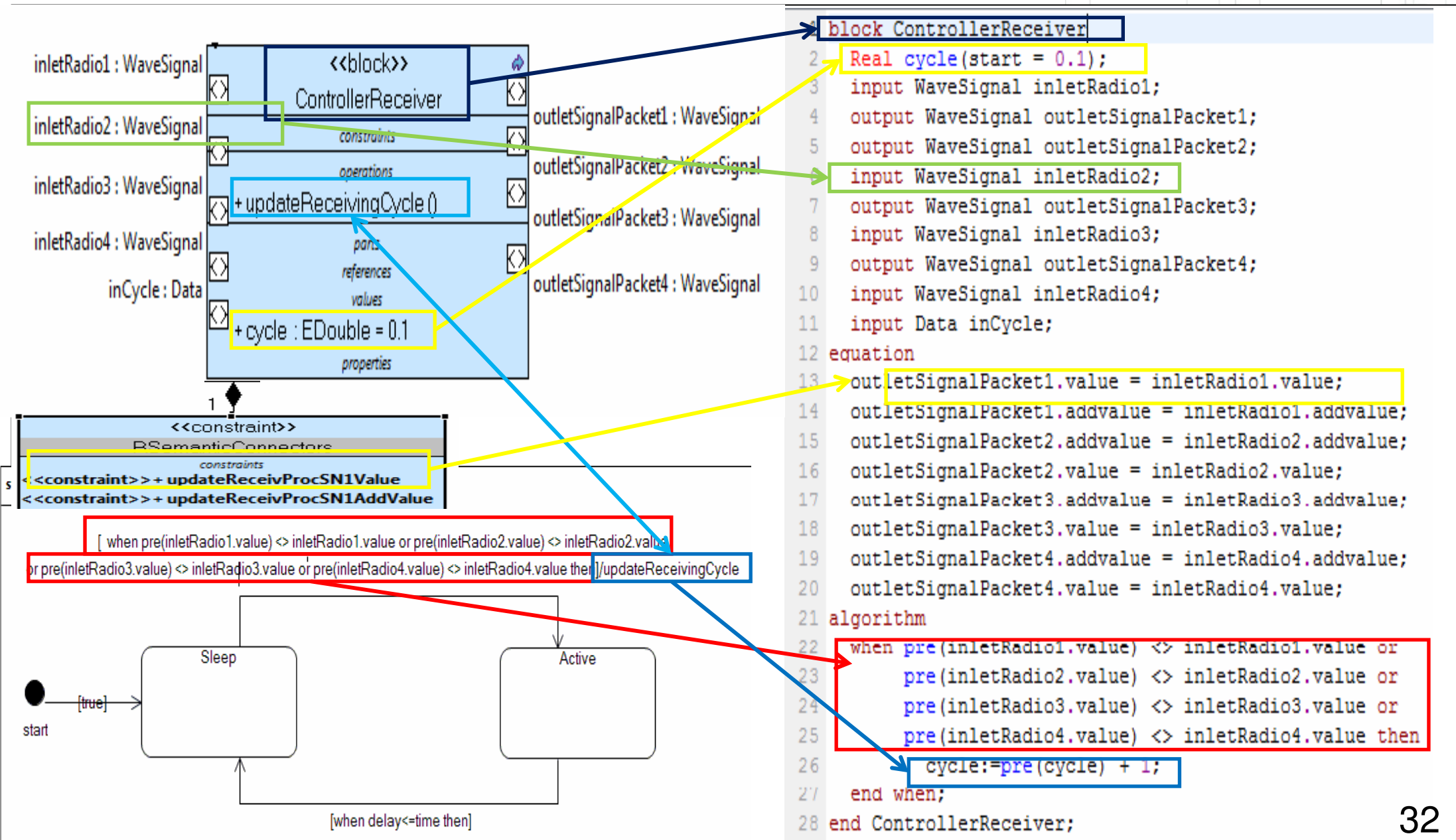
Architecture of a node



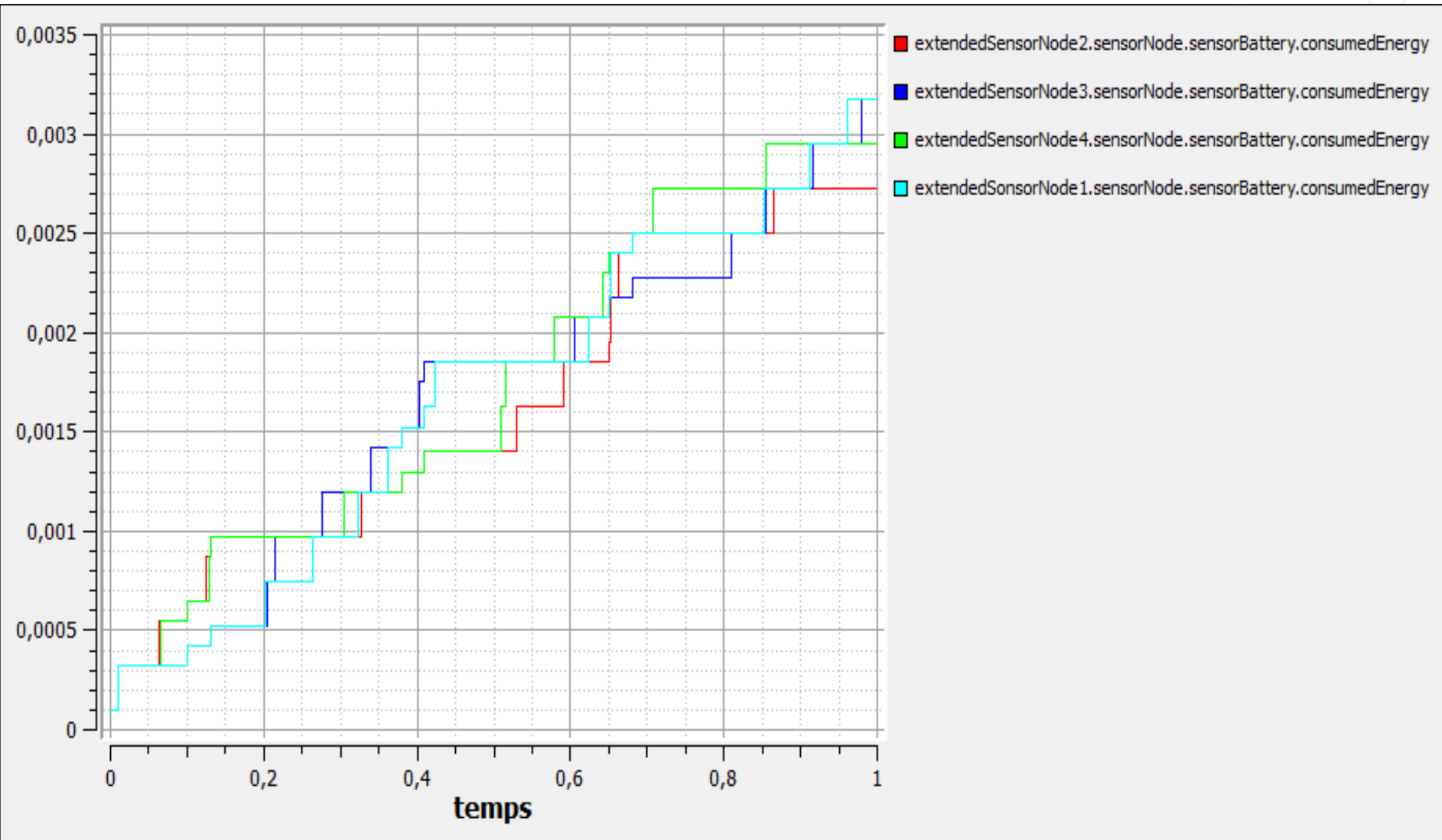
Parametric Diagram



SysML2Modelica



Consumption Energy with a sensor



01 unité de temps : représente 100 seconds.